Unit 3 Study Guide

Unit 3 Energy and Matter · Short on time? Do the starred questions first. Revised November 2026.

Name _____ Period ___ Date _____

LT 3.1 · ATP: how a cell spends and rebuilds energy

1 Draw ATP. Label adenine, ribose and the three phosphate groups. Circle the phosphate that comes off to make ADP.

2 When a cell spends ATP, what happens to the molecule? When the cell rebuilds ATP, what gets added, and is energy stored or released?

3 Write *immediate*, *short-term* or *long-term* beside each. ATP _____ carbohydrates _____ lipids _____

4 What type of macromolecule is ATP?

5 Why do people compare ATP and ADP to a rechargeable battery?

6 ★ A cell holds only a small supply of ATP at any moment. Where does the energy come from to keep rebuilding it?

LT 3.2 · Photosynthesis: light to sugar

7 Write the word equation and the chemical equation for photosynthesis. Label the reactants and the products.

8 What is the purpose of photosynthesis? Finish the sentence: _____ energy is turned into _____ energy.

9 ★ Which organisms carry out photosynthesis? Cyanobacteria have no chloroplasts, so where does their photosynthesis happen?

10 Name the organelle and the pigment that capture light. Inside the chloroplast, which part is the stacks of discs, and which is the fluid around them?

11 ★ (a) As carbon dioxide increases, what happens to the rate of photosynthesis, and why does the line level off? (b) Why does the rate fall at the highest temperatures?

12 ★ What moves through the stomata of a leaf? What happens to the rate of photosynthesis when the stomata close, and why?

LT 3.3 · Cellular respiration, with and without oxygen

13 Write the word equation and the chemical equation for aerobic cellular respiration.

14 What is the purpose of cellular respiration?

15 ★ Which carry out cellular respiration: autotrophs, heterotrophs or both? How is cellular respiration different from breathing?

16 ★ Eukaryotes run aerobic respiration in which organelle? Some prokaryotes use oxygen too and have no mitochondria. Where do they make ATP?

17 Compare the two kinds of respiration. Aerobic and anaerobic: Uses oxygen? ATP per glucose? Where in the cell?

18 What is another name for anaerobic respiration? Which kind happens in tired muscles? Which kind makes bread rise?

LT 3.4 · How the two processes connect

19 ★ What does each process use that the other one makes?

20 Draw a leaf cell with one chloroplast and one mitochondrion. Add arrows for sunlight, carbon dioxide, water, glucose, oxygen, ATP and heat.

21 ★ Count the atoms on each side of the photosynthesis equation. Why do the two sides have to match?

22 ★ Does a plant carry out photosynthesis in the dark? Cellular respiration? What happens to the carbon dioxide around a plant at night?

23 ★ Trace the energy from the sun to a moving muscle. Why do we say matter cycles and energy flows?

24 What is metabolism? Are photosynthesis and cellular respiration part of it? Explain.